WACV 2027 UPAR Challenge
Fact sheet

This is the fact sheet's template for the WACV 2027 UPAR Challenge. Please fill out the following sections carefully in a scientific writing style. Then, send the compressed project (in `.zip` format), i.e., the generated PDF, `.tex`, `.bib` and any additional files to `upar.challenge@gmail.com`.

I. TEAM DETAILS

- Team leader name: Kaveh Safavigerdini
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- Team leader affiliation: Electrical Engineering and Computer Science Department, University of Missouri - Columbia, Mo, USA
- Team leader email: ksgh2@umsystem.edu
- Name of other team members (and affiliation): —
- Team website URL (if any): —
- Competition track (mark with X one single option)[1]:
  - (X) Track 1: Pedestrian Attribute Recognition.
  - ( ) Track 2: Attribute-based Person Retrieval.
  - ( ) Track 3: Human Pose Estimation.

II. CONTRIBUTION DETAILS

A. Title of the contribution

Fine-Tuning a CLIP ViT-B/16 Visual Encoder with Tempered Class-Balanced BCE and Coordinate-Ascent Logit Calibration for Unified Pedestrian Attribute Recognition.

B. Representative image / workflow diagram of the method

A pedestrian crop is letterboxed to 224×224 with grey (128) padding and CLIP-normalized, then encoded by the fine-tuned CLIP ViT-B/16 visual tower [1] (12 blocks, width 768, QuickGELU, class-token pooling, 512-d projection) and a linear head ($\mathbb{R}^{512} \to \mathbb{R}^{40}$) into raw attribute logits $\mathbf{z} \in \mathbb{R}^{40}$. Adding a validation-calibrated bias $\mathbf{b} \in \mathbb{R}^{40}$ yields final probabilities $\hat{\mathbf{p}} = \sigma(\mathbf{z} + \mathbf{b})$ for the challenge's fixed 0.5 threshold.

C. Detailed method description

Architecture. We adopt the visual tower of OpenAI CLIP ViT-B/16 (16×16 patches, 12 pre-norm Transformer blocks of width 768 with 12 heads and QuickGELU, class-token pooling, and a 768×512 projection) initialized from public `openai` weights [1], topped with a single linear layer ($\mathbb{R}^{512} \to \mathbb{R}^{40}$) for the 40 UPAR attributes [2], [3]. The submission is implemented in pure PyTorch and Pillow with no external dependencies.

[1]If you participated in more than one competition track, you need to share with the organizers one fact sheet per track.

Training objective. To counter extreme class imbalance on the training split (positive rates p_a span 0.17% for `LowerBody-Color-Purple` to 93.47% for `Age-Adult`), we use binary cross-entropy with tempered positive-class weights:

$$w_a = \left(\frac{1 - p_a}{\max(p_a,\, 10^{-6})} \right)^{0.5}, \qquad a \in \{1, \ldots, 40\}, \quad (1)$$

compressing weights to $[0.26,\, 24.16]$ to boost rare-attribute recall without gradient instability.

Optimization & data. We train exclusively on the UPAR training split (97,669 images: 10,000 Market-1501, 79,001 PA-100K, 8,668 PETA) for 6 epochs with batch size 256 (`drop_last=True`), AdamW (weight decay 0.05), differential learning rates (2×10^{-5} backbone, 1×10^{-3} head), 500-step linear warmup plus cosine decay, bf16 AMP, and gradient-norm clipping at 1.0. The validation split (33,407 images) is used solely to select the best checkpoint (epoch 2, raw validation Challenge_Avg $= 0.8751$) and calibrate logit biases.

Post-processing (coordinate-ascent logit calibration). Since evaluation enforces a fixed 0.5 threshold while ranking by Challenge_Avg (harmonic mean of mA and Instance_F1), we tune a 40-d logit bias $\mathbf{b} \in [-3, 3]^{40}$ (and temperature $T \in \{1, 2\}$; $T = 1$ optimal) on the validation set via 2 coordinate-ascent sweeps over a 17-point grid per attribute (step 0.375). Predicting $\sigma(\mathbf{z} + \mathbf{b})$ implements per-attribute thresholds $\tau_a = \sigma(-b_a)$ under the 0.5 rule (32/40 biases non-zero), lifting validation Challenge_Avg from 0.8751 to 0.8835 (mA: 0.8717 $\to$ 0.8941; Instance_F1: 0.8784 $\to$ 0.8731).

D. Challenge results

Validation (33,407 images) and official test leaderboard (36,343 images) results:

Metric	Val (raw)	Val (cal.)	**Test (leaderboard)**
Challenge_Avg	0.8751	0.8835	**0.7599**
mA	0.8717	0.8941	0.7716
Label_F1	0.7049	0.6673	0.5341
Instance_Acc	0.8001	0.7913	0.6257
Instance_Prec	0.8690	0.8494	0.7152
Instance_Rec	0.8953	0.9070	0.8009
Instance_F1	0.8784	0.8731	0.7486

E. Final remarks

Pros. (i) Fine-tuning a CLIP ViT-B/16 visual tower with a single linear head gives strong cross-domain features without part-localization branches. (ii) Tempered BCE weighting

($\gamma = 0.5$) and coordinate-ascent logit calibration directly align predictions with the harmonic Challenge_Avg metric. (iii) Self-contained pure-PyTorch inference with batched bf16 execution.

Cons. (i) The validation-to-test drop ($0.8835 \rightarrow 0.7599$) reflects domain shift and mild bias-calibration overfitting to validation priors (trading Label_F1 precision for mA recall). (ii) No ensembling or test-time augmentation. (iii) Rare color attributes remain the main error source.

III. Additional method details

Please, reply if your challenge entry considered (or not) the following strategies and provide a brief explanation. For each question, mark with X one single option.

- **For the competition track associated with this fact sheet, you confirm that you have trained your model on the predefined datasets** (X) Yes, ( ) No
  Trained strictly on the predefined UPAR training split (97,669 images: Market-1501, PA-100K, PETA).
- **Did you use any pre-trained model:** (X) Yes, ( ) No.
  If yes, please detail (e.g., model architecture, training strategy, train data, etc.).
  Initialized from the OpenAI CLIP ViT-B/16 visual tower (`openai` weights via `open_clip_torch`, WIT-400M [1]), fine-tuned end-to-end with a linear head ($\mathbb{R}^{512} \rightarrow \mathbb{R}^{40}$).
- **Did you use external data?** ( ) Yes, (X) No
  If yes, please detail:
  No external images, annotations, or pseudo-labels beyond the public CLIP initialization weights.
- **Did you perform any data augmentation?**
  (X) Yes, ( ) No
  If yes, please detail:
  Random horizontal flip ($p = 0.5$) and tensor-space random erasing ($p = 0.25$, rectangle size $\in [1/8, 1/3)$ per side filled with 0.0, the normalized CLIP mean).
- **Did you use the provided validation set as part of your training set?** ( ) Yes, (X) No
  If yes, please detail:
  Never used for gradient updates; used only for epoch-2 checkpoint selection and post-hoc logit-bias calibration.
- **Did you use any regularization strategies/terms?**
  (X) Yes, ( ) No
  If yes, please detail:
  AdamW weight decay (0.05), gradient ℓ_2-norm clipping at 1.0, random erasing ($p = 0.25$), and early stopping at epoch 2.
- **Did you use any pre-processing steps?** (X) Yes, ( ) No
  If yes, please detail:
  Aspect-ratio-preserving bilinear letterbox resize to 224×224 with grey (128) padding and CLIP normalization ($\mu = (0.4815, 0.4578, 0.4082)$, $\sigma = (0.2686, 0.2613, 0.2758)$).
- **Did you use any post-processing steps?** (X) Yes, ( ) No
  If yes, please detail:
  Per-attribute logit bias $\mathbf{b} \in [-3, 3]^{40}$ tuned via 2-sweep coordinate ascent (17-point grid, $T = 1.0$) on validation Challenge_Avg; inference outputs $\sigma(\mathbf{z} + \mathbf{b})$ thresholded at 0.5.
- **Did you use Vision Language Models (VLMs) of any kind?** (X) Yes, ( ) No
  If yes, please detail:
  Only the visual tower of CLIP ViT-B/16 [1] is used as an image backbone; the text encoder and language supervision are unused.

IV. Code repository

Link to a code repository with complete and detailed instructions so that the results obtained on Codabench can be reproduced locally. This includes a list of requirements, pre-trained models, and so on. Note, training code with instructions is also required. This is required for all participants to appear in the final ranking. **Organizers strongly encourage the use of docker to facilitate reproducibility**.

Code repository: `https://github.com/kavehsfv/UPAR_Track1`

References

[1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proceedings of the 48th International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.

[2] A. Specker, M. Cormier, and J. Beyerer, "UPAR: Unified pedestrian attribute recognition and person retrieval," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, January 2023, pp. 981–990.

[3] M. Cormier, A. Specker, J. C. S. J. Junior, L. Moritz, J. Metzler, T. B. Moeslund, K. Nasrollahi, S. Escalera, and J. Beyerer, "UPAR challenge 2024: Pedestrian attribute recognition and attribute-based person retrieval – dataset, design, and results," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*, 2024, pp. 359–367.